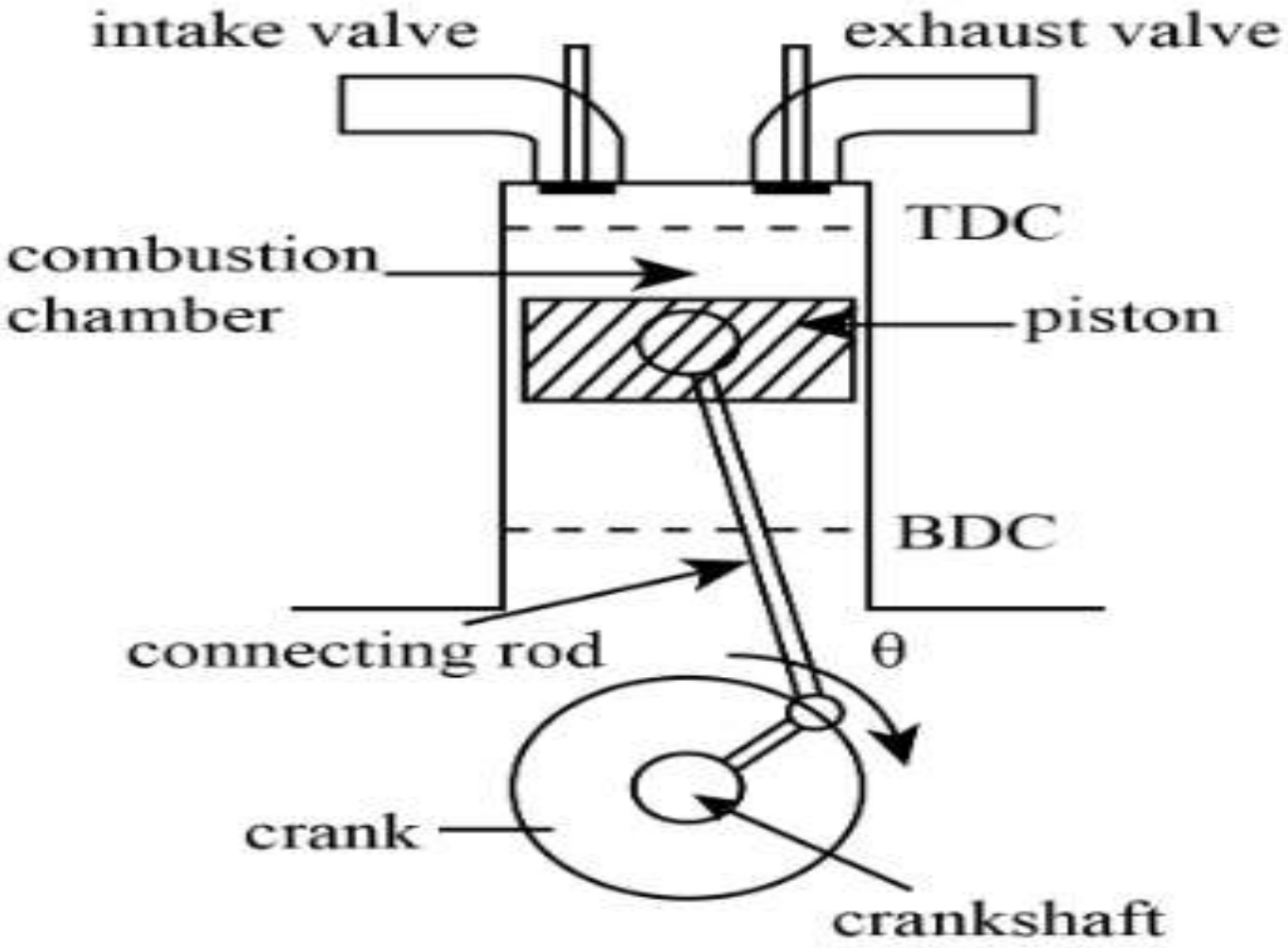
MODULE-II
IC Engines

Introduction, petrol engine, diesel engines,
Working of four Stroke engines
Applications,
simple numerical.

Introduction to IC Engines

- The term "IC engine" signifies its meaning as an internal combustion engine.
- It is type of heat engine which convert the heat energy released during the combustion of fuel into mechanical work.
- The combustion process in an IC engine takes place in the internal part of the engine that is known as the combustion chamber.
- Example- diesel engine, petrol engine, etc.

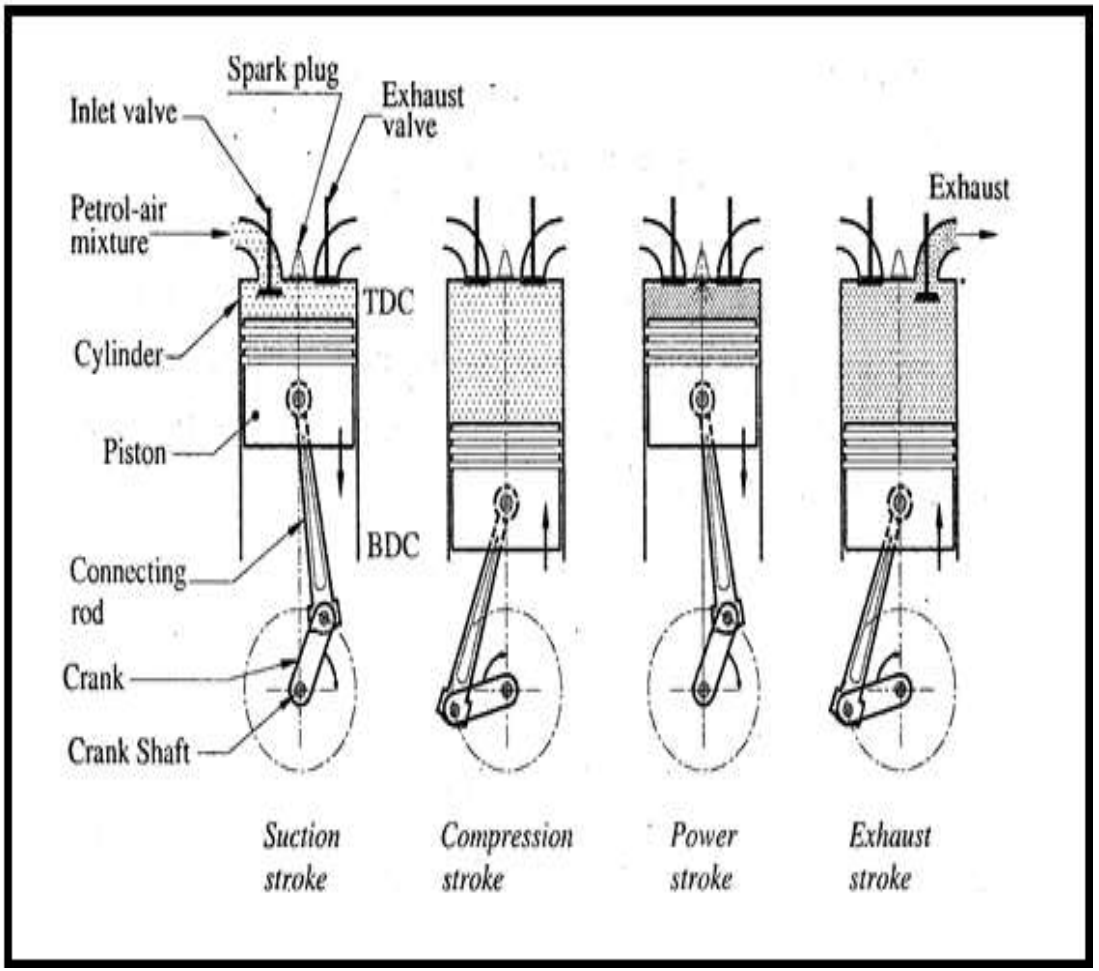
Components of IC engine

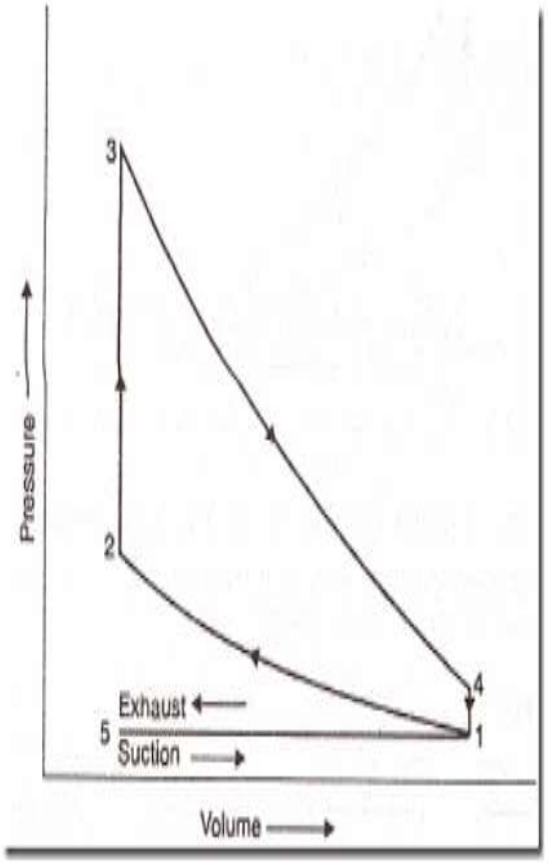


Components of IC engine

- **Intake valve:** A valve that opens to allow the fresh air-fuel mixture to enter the combustion chamber during the intake stroke.
- **Exhaust valve:** A valve that opens to allow the exhaust gases to exit the combustion chamber during the exhaust stroke.
- **Cylinder (combustion chamber):** It is considered as heart of an engine in which combustion of fuel take place. The inside diameter of cylinder is called as bore
- **Piston:** A cylindrical component that moves up and down inside the cylinder, driven by the force generated by the combustion process.
- **Connecting rod:** Connects the piston to the crankshaft and transfers the linear motion of the piston into rotational motion.
- **Crank &Crankshaft:** A crank is lever that is connected to the end of connecting rod by pin joint with its other end connected to a shaft called crankshaft.
- Crankshaft Converts the reciprocating motion of the pistons into rotational motion, which drives the transmission and, ultimately, the wheels.
- **TDS (top dead center):** The position of the piston when its at the top most position.
- **BDC (bottom dead center):** The position of the piston when its at the lowest position.

four stroke petrol engine diagram





Theoretical P-V diagram of a four-stroke engine

four stroke petrol engine

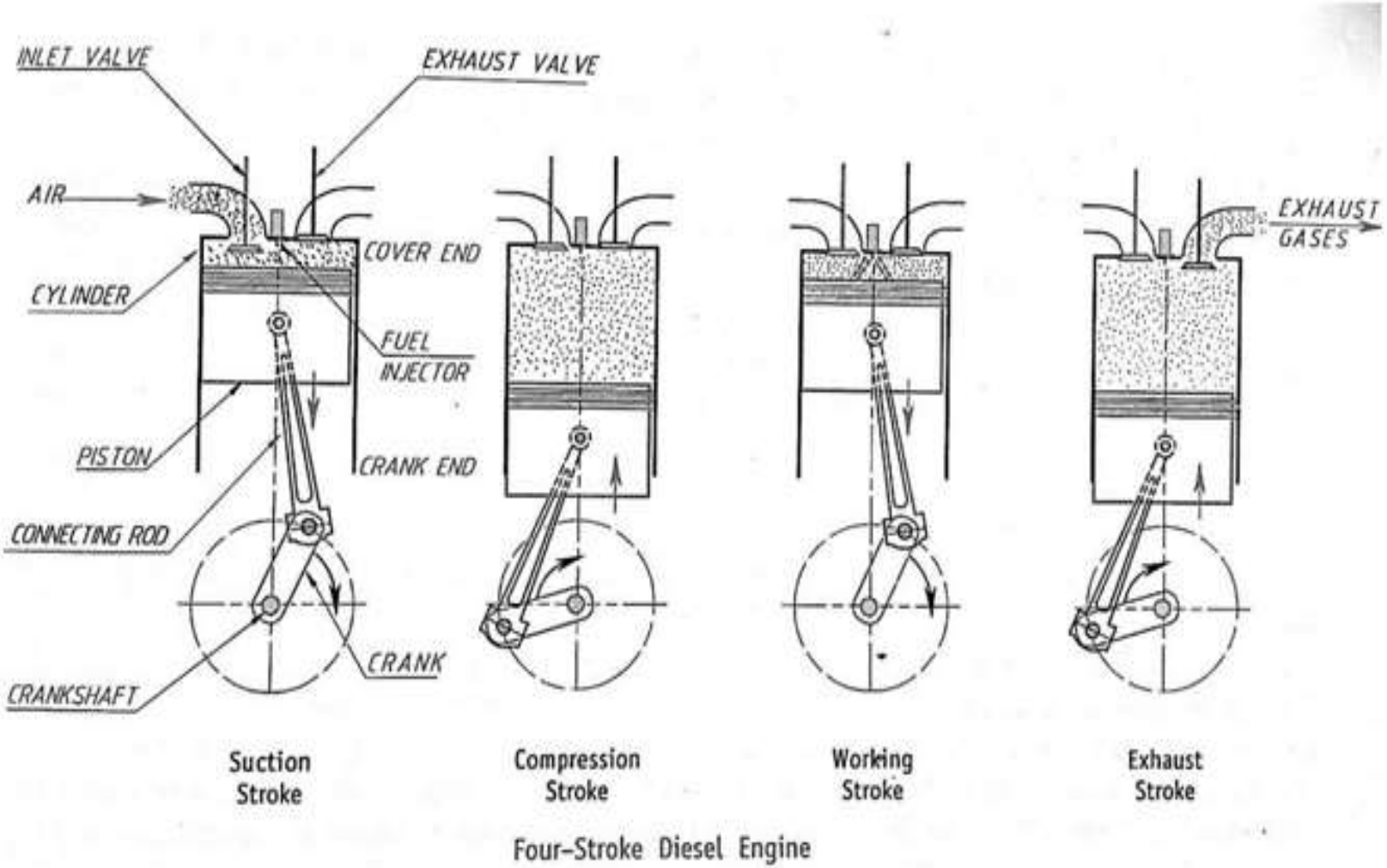
I. **SUCTION STROKE:** In this Stroke the inlet valve opens and proportionate fuel-air mixture is sucked in the engine cylinder. Thus the piston moves from top dead centre (T.D.C.) to bottom dead centre (B.D.C.). The exhaust valve remains closed through out the stroke. This operation is represented by the line 5-1.

II. **COMPRESSION STROKE:** In this stroke both the inlet and exhaust valves remain closed during the stroke. The piston moves towards (T.D.C.) and compresses the enclosed fuel-air mixture drawn. Just before the end of this stroke the operating plug initiates a spark which ignites the mixture and combustion takes place at constant pressure. This operation is represented by 1-2.

III. **POWER STROKE OR EXPANSION STROKE:** In this stroke both the valves remain closed during the start of this stroke but when the piston just reaches the B.D.C. the exhaust valve opens. When the mixture is ignited by the spark plug the hot gases are produced which drive or throw the piston from T.D.C. to B.D.C. and thus the work is obtained in this stroke. This operation is represented by 2-3.

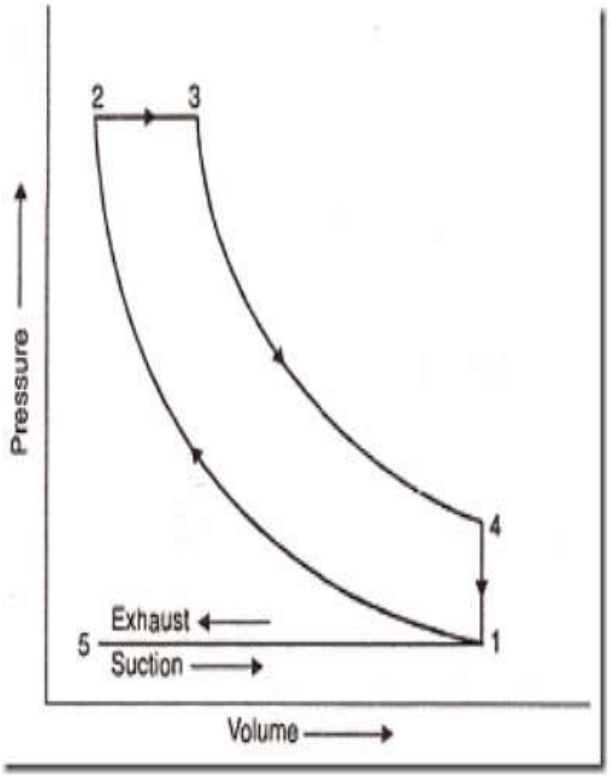
IV. **EXHAUST STROKE:** This is the last stroke of the cycle. Here the gases from which the work has been collected become useless after the completion of the expansion stroke and are made to escape through exhaust valve to the atmosphere. This removal of gas is accomplished during this stroke. The piston moves from B.D.C. to T.D.C. and the exhaust gases are driven out of the engine cylinder. This stroke is represented by the line 1-5.

Four Stroke Diesel engine



Four Stroke Diesel engine

- **SUCTION STROKE:** With the movement of the piston from T.D.C. to B.D.C. during this stroke, the inlet valve opens and the air at atmospheric pressure is drawn inside the engine cylinder; the exhaust valve however remains closed. This operation is represented by the line 5-1.
- **COMPRESSION STROKE:** The air drawn at atmospheric pressure during the suction stroke is compressed to high pressure and temperature as the piston moves from B.D.C. to T.D.C. Both the inlet and exhaust valves do not open during any part of this stroke. This operation is represented by 1-2.
- **POWER STROKE OR STROKE:** As the piston starts moving from T.D.C to B.D.C, the quantity of fuel is injected into the hot compressed air in fine sprays by the fuel injector and it (fuel) starts burning at constant pressure shown by the line 2-3.
- **EXHAUST STROKE:** The piston moves from the B.D.C. to T.D.C. and the exhaust gases escape to the atmosphere through the exhaust valve. When the piston reaches the T.D.C. the exhaust valve closes and the cycle is completed. This stroke is represented by the line 1-5.



Theoretical p- *V* diagram of a four-stroke Diesel Engine

Application of IC Engines

1) Automotive Vehicles: The major application of IC Engine is in cars, motorcycles, trucks, buses, and other vehicles.
2) Aircraft: Another application of IC Engine is to power various types of aircraft, including general aviation planes and some helicopters.
3) Marine Vessels: IC engines are used in boats, ships, and other watercraft for propulsion.
4) Power Generators: IC engines are used in generators to produce electricity for various applications, especially in remote areas and as backup power sources.
5) Construction Equipment: IC engines are found in various construction machinery like excavators, bulldozers, and cranes.
6) Agriculture: Tractors and other farming equipment often use IC engines to perform various tasks on farms.
7) Industrial Machinery: IC engines are used in various industrial machines, including compressors, pumps, and generators.

Performance of IC engines

1) **Indicated Power** :- In four stroke I C engines, one cycle will be completed in two revolutions of the crank shaft. Therefore the work will be produced in every alternate revolutions of the crankshaft.

- The number of cycles per minute will be equal to half the number of revolutions per minute.
- IP= Pm × L × A × N / 60 × 2× 1000 Kw (when Pm is expressed in N/m^2)
- IP= 100 x Pm × L× A × N / 60 × 2 Kw (when Pm is expressed in bar)
- Where:
- IP = Indicated Power (in kilowatts)
- Pm = Mean effective pressure (in bar or N/m^2)
- L = Stroke length (in meters, m)
- A = Area of the piston (in m^2)
- N = RPM of the crankshaft
- n= number of cycles per minute

2) **Brake power** :- the net power available at the crankshaft will be equal to the difference between the indicated power produced inside the engine cylinder and power lost due to friction.

- BP = (2π × N × T) / 60
- Where:
- BP = Brake Power (kW)
- N = RPM of the crankshaft
- T = Torque (in K.Nm) = 9.81 × w × R / 1000
- W = net load acting on the brake drum, Kg
- R = radius of the brake drum, m
- 2π = Converts rotational motion to power

Performance of IC engines

3) **Mechanical efficiency** :- it is defined as the ration of the brake power and the indicated power. It is expressed in percentage.

- η mech = (I P / B P) × 100
- Where:
- ηmech = Mechanical efficiency (in %)
- BP = Brake Power (actual output power at the crankshaft)
- IP = Indicated Power (power developed in the combustion chamber)

4) **Thermal efficiency** :- it is defined as the ratio of the power developed by the engine to the heat supplied by the fuel in the same interval of time. It is expressed in percentage.

- η thermal = (Power Output / Heat energy Supplied by the fuel) ×100
- For engines, thermal efficiency can be expressed more specifically as:
- **a. Indicated Thermal Efficiency**:
- I thermal = (indicated power / Heat energy Supplied by the fuel) ×100
- I thermal = (I P / CV × m) × 100
- **b. Brake Thermal Efficiency**:
- B thermal = (brake power / Heat energy Supplied by the fuel) ×100
- B thermal = (B P / CV × m) × 100
- Where:
- IP = Indicated Power (kW)
- BP = Brake Power (kW)
- m= Mass flow rate of fuel (kg/s)
- CV = Calorific value of the fuel (kJ/kg)

Simple problems

1. A four stroke IC engine running at 450 rpm has a bore diameter of 100mm and stoke length 120mm. The indicator diagram details are ; area of diagram 4 cm^2 , length of indicator diagram 6.5 cm and the spring value of the spring used is 10 bar/cm. calculate the indicated power of the engine.
2. A four stroke single cylinder IC engine of 250mm cylinder diameter and 400mm stroke runs at a piston speed of 8 m/s. if the engine develops 50kw indicated power, find its mean effective pressure and the crankshaft speed.
3. A four stoke petrol engine of 100mm bore and 150mm stroke consumes 1kg of fuel per hour. The mean effective pressure is 7 bar and its indicated thermal efficiency is 30%. The calorific value of the is 40 x 10^3 KJ/Kg. find the crankshaft speed.
4. The following data refers to a single cylinder 4 stroke petrol engine.

a) Cylinder diameter = 20cm
b) Stroke of the piston = 40cm
c) Engine speed = 400 rpm
d) Indicated mean effective pressure = 7 bar
e) Fuel consumption = 10 liters / hour
f) Calorific value of the fuel = 45000 kJ/kg
g) Specific gravity of the fuel = 0.8

Find the indicated thermal efficiency.

Simple problems

5. On a single cylinder four stroke petrol engine, the following readings were taken :

a) Load on a brake drum = 40kg
b) Spring balance reading = 5kg
c) Diameter of the brake drum = 120cm
d) Fuel consumption = 3 kg/hour
e) Calorific value of the fuel = 42000 kj/kg
f) Engine speed = 500 rpm

Find the brake thermal efficiency.

6. The following observations were obtained during a trial on a four stroke diesel engine.

a) Cylinder diameter = 25 cm
b) Stroke of the piston = 40 cm
c) Crankshaft speed = 250 rpm
d) Brake load = 70kg
e) Brake drum diameter = 2m
f) Mean effective pressure = 6 bar
g) Diesel oil consumption = 0.1 m/min
h) Specific gravity of diesel = 0.78
i) Calorific value of diesel = 43900 kj/kg

Find: brake power, indicated power, frictional power, mechanical efficiency , brake thermal efficiency and indicated thermal efficiency.

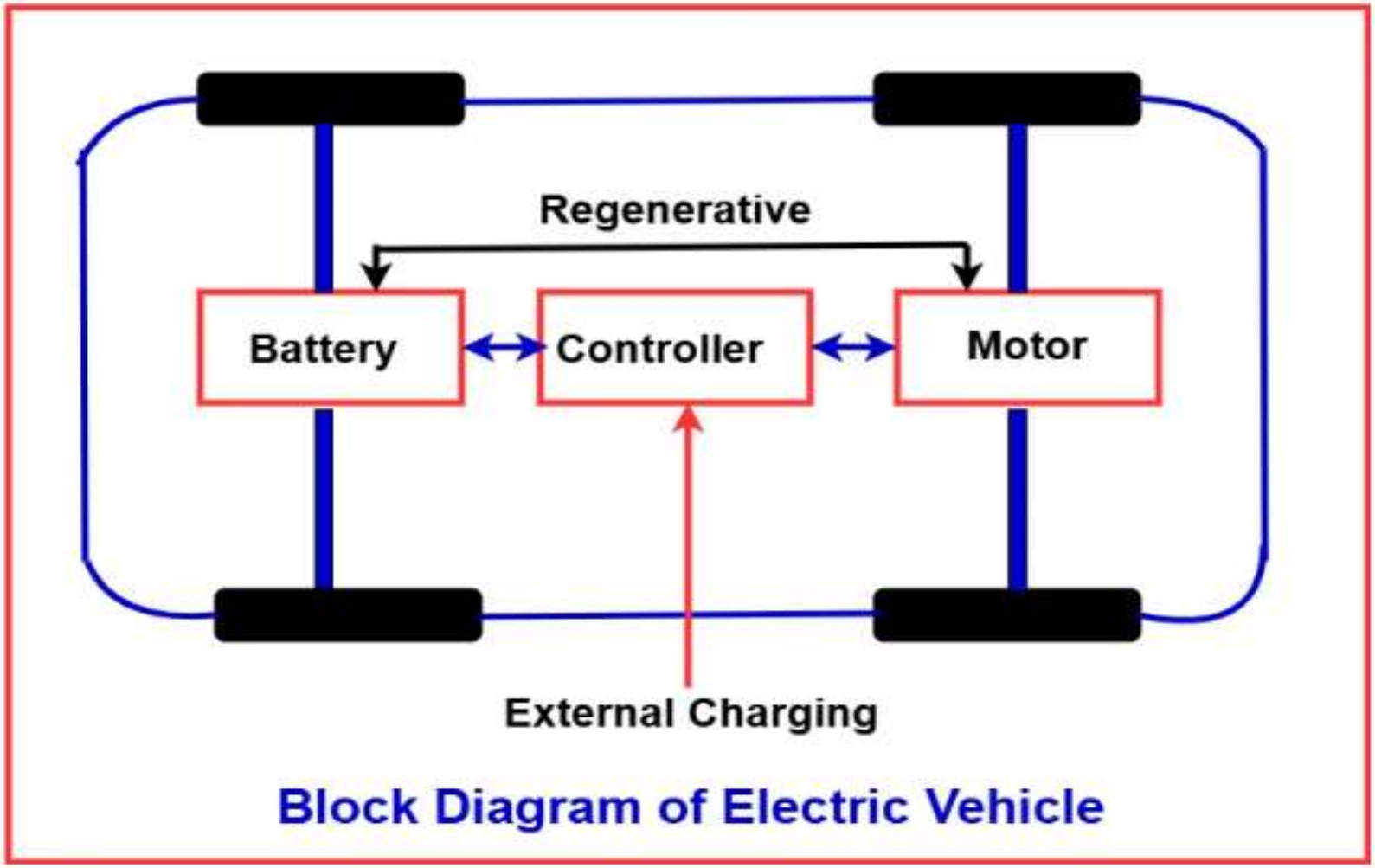
Insight into Future Mobility:
Electric and Hybrid Vehicles

- The future of mobility is transforming towards cleaner, smarter, and more efficient transportation.
- Electric Vehicles (EVs) and Hybrid Electric Vehicles (HEVs) are key innovations addressing environmental concerns, fuel dependency, and urban sustainability.

Electric Vehicles (EVs)

- An **Electric Vehicle (EV)** is a type of vehicle that is powered by electricity instead of traditional internal combustion engines (ICEs) that use petrol or diesel.
- EVs use an electric motor, battery pack, and electronic controllers to generate and regulate power.
- They are considered an eco-friendly alternative to conventional vehicles due to their zero-emission nature and energy efficiency.

Block Diagram of Electric Vehicle



Components of Electric Vehicle

- **Battery Pack** – Stores electrical energy and supplies power to the system.
- **Power Converter (DC-AC Converter)** – Regulates voltage levels for different subsystems.
- **Electric Motor Controller** – Manages power flow from the battery to the motor.
- **Electric Motor** – Converts electrical energy into mechanical energy to drive the wheels.
- **Transmission System** – Transfers power from the motor to the wheels efficiently.
- **Regenerative Braking System** – Converts kinetic energy back into electrical energy during braking.
- **Charging System** – Recharges the battery using external power sources or regenerative braking.

Advantages and disadvantages of Electric Vehicles

Advantages

1) **Zero Emissions**: No emissions, reducing air pollution.
2) **Lower Operating Costs**: Electricity is cheaper than gasoline or diesel, reducing fuel costs.
3) **Smooth and Silent Operation**: Minimal noise and vibration compared to ICE vehicles.
4) **Regenerative Braking**: Enhances efficiency by recovering energy during braking.
5) **Renewable Energy Usage**: Can be powered by clean energy sources like solar or wind.

Disadvantages

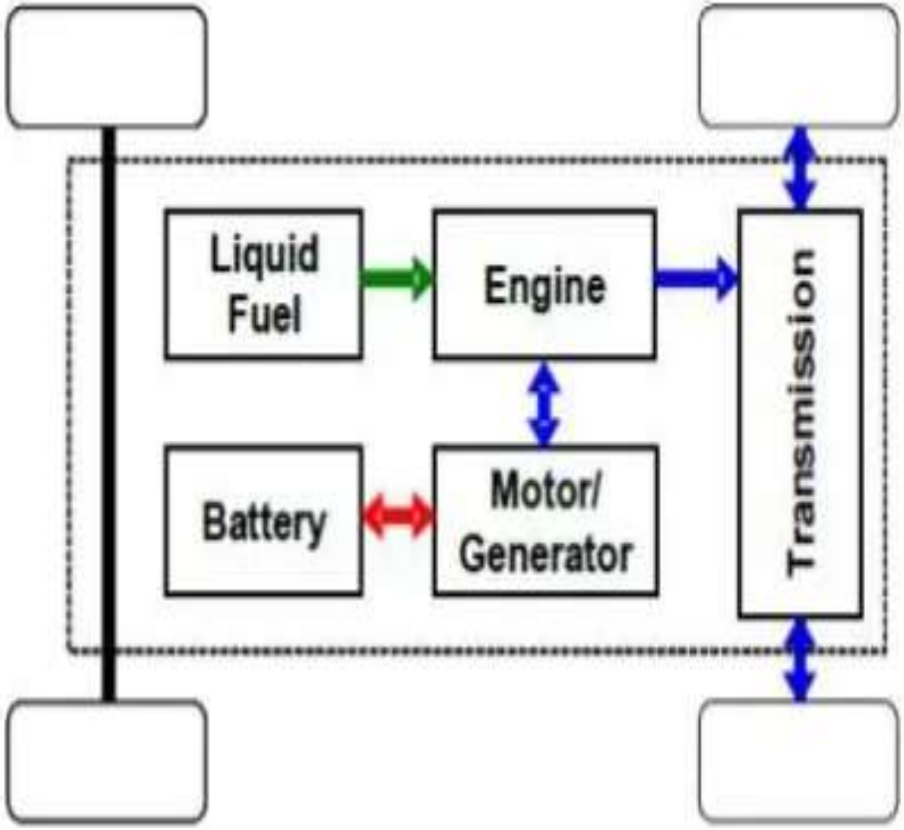
1) **High Initial Cost**: EVs are more expensive upfront due to battery technology.
2) **Limited Range**: Most EVs have a limited driving range per charge compared to fuel-based cars.
3) **Charging Infrastructure**: Availability of charging stations is still a concern in many areas.
4) **Long Charging Time**: Charging an EV takes longer compared to refueling a gasoline vehicle.
5) **Battery Degradation**: Over time, batteries lose efficiency and require costly replacements.

Hybrid Electric Vehicles (HEVs)

- A Hybrid Electric Vehicle (HEV) is a vehicle that combines a conventional internal combustion engine (running on petrol or diesel) with an electric motor powered by a battery, for better efficiency and lower pollution.

Major Components of HEV

1) Internal Combustion Engine (ICE)
2) Electric Motor
3) Battery Pack
4) Power Control Unit (PCU)
5) Transmission System
6) Inverter
7) Hybrid Control Unit (HCU)
8) Regenerative Braking System
9) Generator



Major Components of HEV

1. Internal Combustion Engine (ICE) :- The internal combustion engine in an HEV operates similarly to that in traditional vehicles. It burns fuel (typically gasoline or diesel) to generate power.
2. Electric Motor :- The electric motor is a crucial component of HEVs, providing additional power to the wheels. It works in conjunction with the ICE to reduce fuel consumption and emissions.
3. Battery Pack :- HEVs feature a high-capacity battery pack that stores energy to power the electric motor. Most HEVs use lithium-ion, nickel-metal hydride, or lead-acid batteries.
4. Power Control Unit (PCU) :- The Power Control Unit manages the flow of energy between the ICE, electric motor, and battery.

5) Transmission System :- The transmission system in HEVs transmits power from the engine and electric motor to the wheels.
6) Inverter :- The inverter converts the DC (direct current) electricity from the battery into AC (alternating current) electricity needed by the electric motor.
7) Hybrid Control Unit (HCU): Acts as the brain of the system, controlling power distribution and efficiency.
8) Regenerative Braking System :- Regenerative braking captures kinetic energy produced during braking and converts it into electrical energy.
9) Generator :- In many HEVs, a generator converts mechanical energy from the ICE or regenerative braking into electrical energy.

Advantages and disadvantages of Hybrid Electric Vehicles

Advantages

1) **Extended Driving Range: HEVs can switch between gasoline and electric power, offering greater range and flexibility.**
2) **Fuel Efficiency: HEVs are more fuel-efficient than traditional gasoline-powered vehicles.**
3) **No Charging Dependency: HEVs do not rely on external charging infrastructure since they generate electricity through regenerative braking and the gasoline engine.**
4) **Lower Emissions: HEVs produce fewer emissions than traditional vehicles.**
5) **Affordable Options: HEVs are often more affordable than Evs.**

Disadvantages

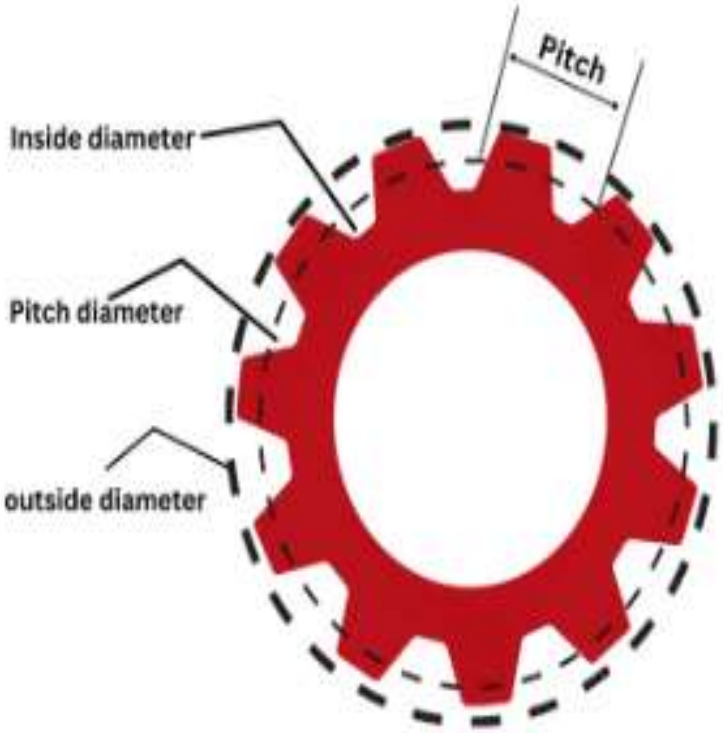
1) **Complexity:** The dual power train increases the complexity of the vehicle, potentially leading to higher maintenance costs.
2) **Environmental Impact:** HEVs still rely on gasoline, which contributes to emissions.
3) **Less Electric Efficiency:** HEVs are not as energy-efficient as EVs since they still rely on fossil fuels.
4) **Performance Limitations:** HEVs often lack the powerful acceleration and smooth performance of EVs.
5) **Battery Size:** HEVs have smaller batteries than EVs, limiting their ability to run solely on electric power for extended periods.

Power Transmission systems Introduction

- A gear is a toothed wheel that engages another toothed mechanism to change speed or the direction of transmitted motion.
- Gears are generally used for different reasons:

1) To increase or decrease the speed of rotation;
2) To change the amount of force or torque;
3) To move rotational motion to a different axis (i.e. parallel, right angles, rotating, linear etc.); and
4) To reverse the direction of rotation.

TYPE OF GEARS : SPUR GEARS



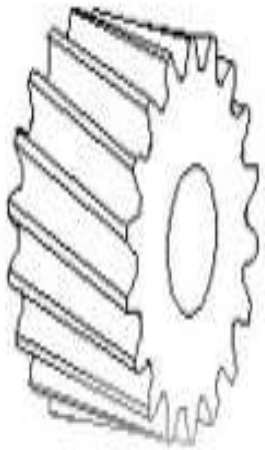


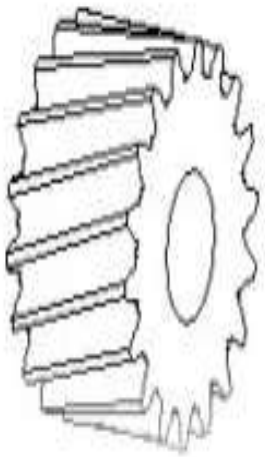
Spur Gears

TYPE OF GEARS : SPUR GEARS

- Spur gears are used to transmit power between two parallel shafts. The teeth on these gears are cut straight and are parallel to the shafts to which they are attached.
- When engaged, spur gears exhibit contact across the entire width parallel to the shaft axis, leading to sudden load application and potentially detrimental impact stresses. This impact loading, combined with the excessive sound at high speeds, makes spur gears more suitable for moderate speeds.

TYPE OF GEARS : HELICAL GEARS







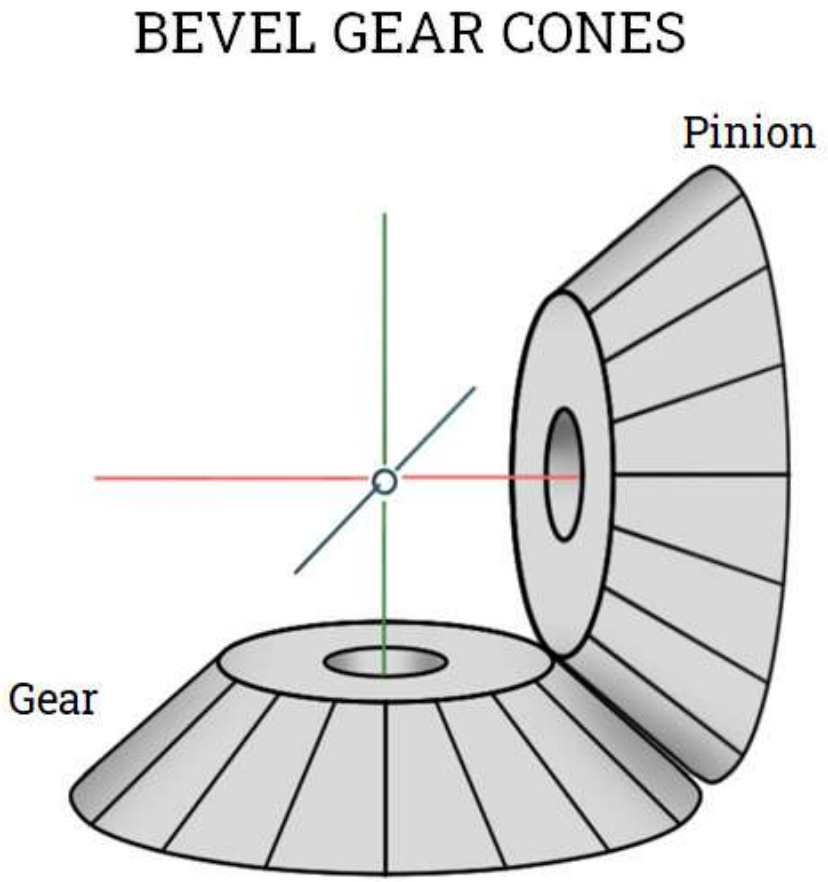


Helical Gears

TYPE OF GEARS : HELICAL GEARS

- Helical gears resemble spur gears, but the teeth are cut at an angle rather than parallel to the shaft axis like on spur gears. The angle that the helical gear tooth is on is referred to as the helix angle.
- The angle of helix depends upon the condition of the shaft design and relative position of the shafts. To ensure that the gears run smoothly, the helix angle should be such that one end of the gear tooth remains in contact until the opposite end of the following gear tooth has found a contact.

TYPE OF GEARS : BEVEL GEARS

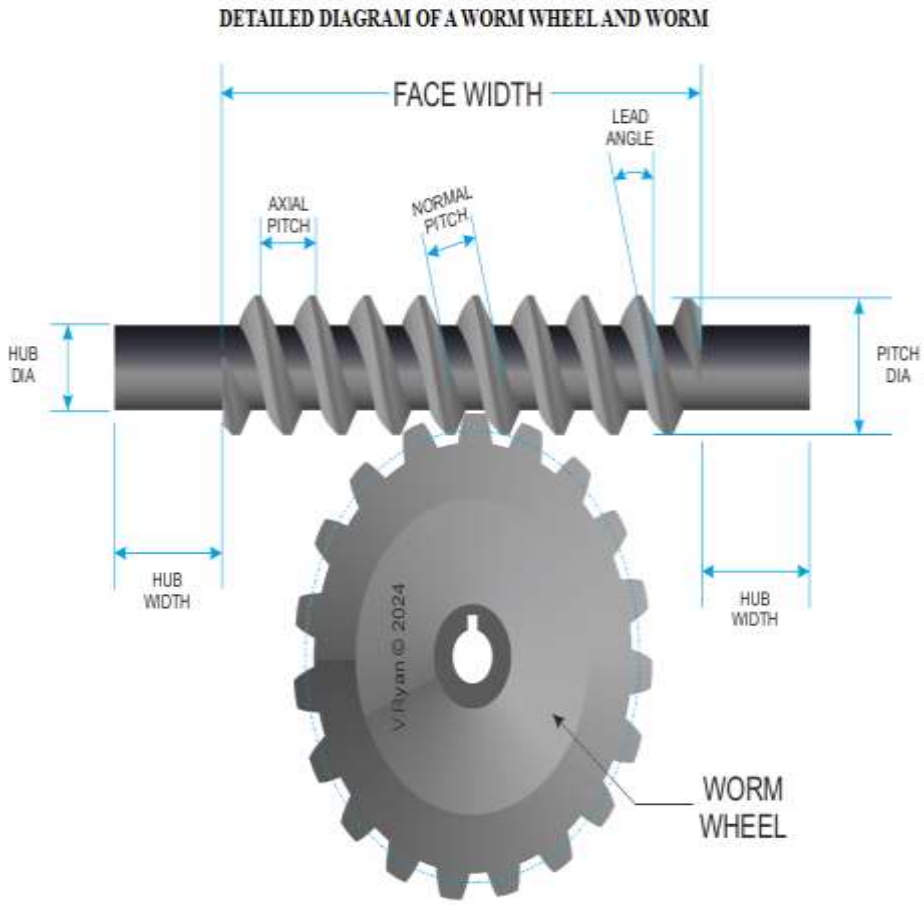




TYPE OF GEARS : BEVEL GEARS

- A bevel gear is shaped like a section of a cone and primarily used to transfer power between intersecting shafts at right angles. The teeth of a bevel gear may be straight or spiral.
- Designed for the efficient transmission of power and motion between intersecting shafts. A good example of bevel gears is seen as the main mechanism for a hand drill. As the handle of the drill is turned in a vertical direction, the bevel gears change the rotation of the chuck to a horizontal rotation.

TYPE OF GEARS : WORM GEARS





Worm Gear

TYPE OF GEARS : WORM GEARS

- Worm gears are used to transmit power between two shafts that are at right angles to each other and are non-intersecting. Worm gears are special gears that resemble screws, and can be used to drive spur gears or helical gears.
- Worm gearing is essentially a special form of helical gearing in which the teeth have line contact and the axes of the driving and driven shafts are usually at right angles and do not intersect.

Velocity ratio of gear drives

- The velocity ratio of a gear drive is defined as the ratio of the speed of the driven gear to the speed of the driving gear.
- Let d1 & d2 be the pitch circle diameters of the driving and driven gear.
- Let T1 & T2 be the number of teeth on the driving and driven gears respectively.
- Let N1 & N2 be their speeds in revolutions per minute.
- Since there is no slip between the pitch cylinders of the two gear wheels, the linear speed of the two pitch cylinders must be equal.
- Therefore :- (linear speed of the pitch cylinder representing the driving gear) = (linear speed of the pitch cylinder representing the driven gear)
- **π d1 N1 = π d2 N2**
- **N1/N2 = d2/d1 ---- equation 1**
- The circular pitch for both the meshing gears remains same**.**
- **Pc = π d1 / T1 = π d2 / T2**
- **d2/ d1 = T2/T1 ------equation 2**
- **From equation 1 & 2 :-** velocity ratio of a gear drive =

  N1/N2 = d2/d1 = T2/T1

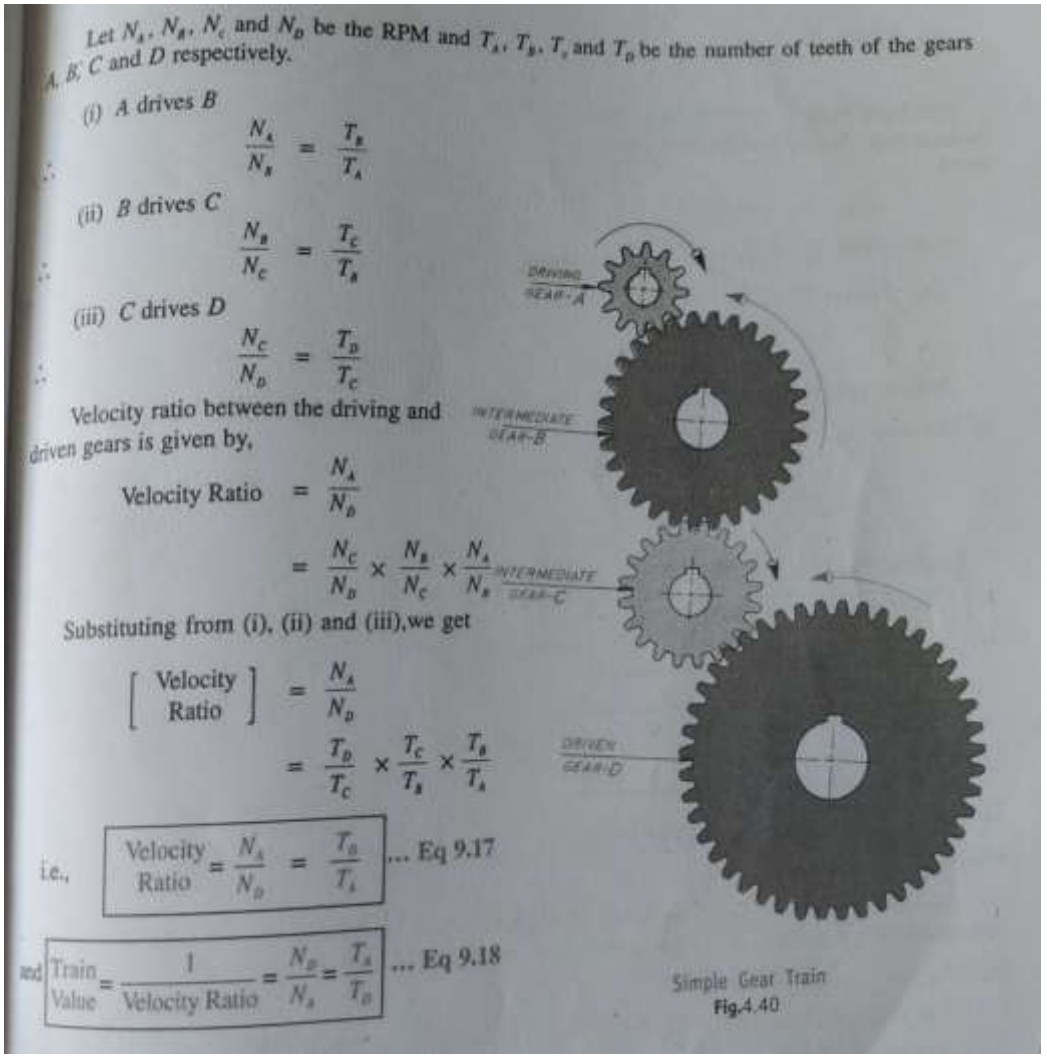
Gear trains

- A gear train is a power transmission system made up of two or more gears, the power can be transmitted between the driving and driven shafts.
- Types of gear trains

1. Simple gear train
2. Compound gear train

1. Simple gear train

- As per figure, A is the driving gear, D is the driven gear and B & C are the intermediate gears, the intermediate gears B & C are also idler gears.
- The idler gears act as the intermediate gears to establish the drive between the driving and the driven gears, but they change the direction of rotation of the driven.



Let N_A, N_B, N_C and N_D be the RPM and T_A, T_B, T_C and T_D be the number of teeth of the gears A, B, C and D respectively.

(i) A drives B

$$\therefore \quad \frac{N_A}{N_B} = \frac{T_B}{T_A}$$

(ii) B drives C

$$\therefore \quad \frac{N_B}{N_C} = \frac{T_C}{T_B}$$

(iii) C drives D

$$\therefore \quad \frac{N_C}{N_D} = \frac{T_D}{T_C}$$

Velocity ratio between the driving and driven gears is given by,

$$\text{Velocity Ratio} = \frac{N_A}{N_D} = \frac{N_C}{N_D} \times \frac{N_B}{N_C} \times \frac{N_A}{N_B}$$

Substituting from (i), (ii) and (iii), we get

$$\left[\begin{matrix}\text{Velocity}\\ \text{Ratio}\end{matrix}\right] = \frac{N_A}{N_D} = \frac{T_D}{T_C} \times \frac{T_C}{T_B} \times \frac{T_B}{T_A}$$

i.e., $\boxed{\text{Velocity Ratio} = \frac{N_A}{N_D} = \frac{T_D}{T_A}}$ ... Eq 9.17

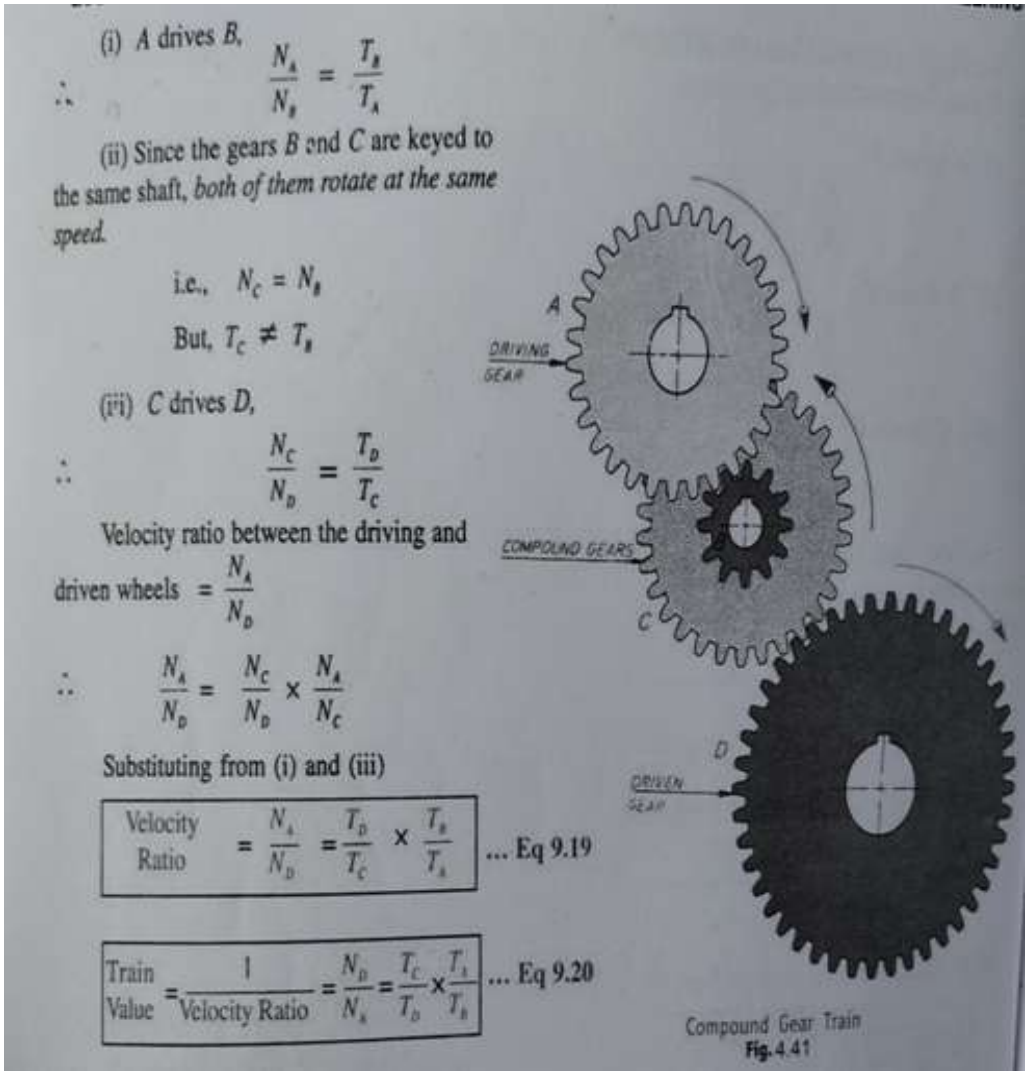
and $\boxed{\text{Train Value} = \frac{1}{\text{Velocity Ratio}} = \frac{N_D}{N_A} = \frac{T_A}{T_D}}$ ... Eq 9.18

Simple Gear Train

Fig.4.40

2. Compound gear train

- In a compound gear train, the intermediate shaft carries two gears which are keyed to it.
- When the velocity ratio is very high, a simple gear train becomes practically impossible owing to difficulty of connecting the driving and driven gears because the given centre distance is small. In such cases a compound gear train is used.
- A is the driving gear, D is the driven gear and B and C keyed to an intermediate shaft and are called compound gears.
- Since the B & C gears are mounted on the same shaft, they rotate at the same speed.



Simple problems on gear drives

1. Two gear wheels having 80 teeth and 30 teeth mesh with each other. If the smaller gear wheel runs at 480 rpm, find the speed of the larger wheel.
2. A gear wheel of 20 teeth drives another gear wheel having 36 teeth running at 200rpm. Find the speed of the driving wheel and the velocity ratio.
3. A compound gear train consists of 4 gears, A.B.C & D and they have 20,30,40, & 60 teeth respectively. A is keyed to the driving shaft, B and C are compound dears, B meshes with A, and C meshes with D. if A rotates at 180 rpm, find the rpm of D.
4. A gear wheel has 50 teeth of module 5mm. Find the pitch diameter and the circular pitch.
5. A simple gear train is made up of four gears A,B,C,& D having 20,40,60 and 70 teeth respectively. If gear A is the main driver rotating at 500rpm clockwise, calculate the following:

   i) speeds of intermediate gears

   ii) speed and direction of the last follower

   iii) train Value